

§7-1 Integration by Parts

Homework. : 3,9,10,15,21,25,29,40,45,57,63

公式 : Integration by parts

- $\int u dv = uv - \int v du .$
- $\int_a^b u dv = uv \Big|_a^b - \int_a^b v du .$

Integration by parts $\leftrightarrow$ Product rule
(積分) (微分)

Why ?

$$\begin{aligned} u &= f(x) \quad v = g(x); \\ (f(x)g(x))' &= f'(x)g(x) + f(x)g'(x) \\ \Rightarrow \int (f(x)g(x))' dx &= \int \underbrace{g(x)}_v \underbrace{f'(x)}_{du} dx + \int \underbrace{f(x)}_u \underbrace{g'(x)}_{du} dx \\ \Rightarrow uv &= \int v du + \int u dv \\ \Rightarrow \int u dv &= uv - \int v du. \end{aligned}$$

Keys :

1. Identify u and dv . Easy to recover v from dv . $\int v du$ is no complicated than $\int u dv$.
2. 可能需要作不只一次的 integration by parts (見 Examples 4.5 and 6) .
3. 可能需要解方程式 (見 Examples 7) .

Example 1 : $\int x \sin x dx = -x \cos x + \int \cos x dx = -x \cos x + \sin x + C.$

Solution :

$$\begin{aligned} u &= x \quad dv = \sin x dx \\ du &= dx \quad v = -\cos x \end{aligned}$$

Example 2 : $\int \tan^{-1} x dx = x \tan^{-1} x - \int \frac{x}{1+x^2} dx .$

Solution :

$$\begin{aligned} u &= \tan^{-1} x \quad dv = dx \\ du &= \frac{dx}{1+x^2} \quad v = x \\ \Rightarrow \int \tan^{-1} x dx &= x \tan^{-1} x - \frac{1}{2} \ln(1+x^2) + C . \end{aligned}$$

Example 3 : $\int \ln(x-1)dx = (x-1)\ln(x-1) - \int 1dx = (x-1)\ln(x-1) - x + C.$

Solution :

$$u = \ln(x-1) \quad dv = dx$$

$$du = \frac{dx}{x-1} \quad v = x-1.$$

Example 4 : $\int x^2 e^x dx.$

Solution :

$$\begin{aligned} \int x^2 e^x dx \\ &= x^2 e^x - \int 2x e^x dx \\ &= x^2 e^x - 2x e^x - 2 \int e^x dx \\ &= x^2 e^x - 2x e^x + 2e^x + C. \end{aligned}$$

$u = x^2$	$dv = e^x dx$
$du = 2x dx$	$v = e^x$
$u = 2x$	$dv = e^x dx$
$du = 2 dx$	$v = e^x$

Example 5 : $\int x^3 e^x dx.$

Solution : 可將上式的 Example 4 的過程作簡化但傳神的設計如左下.

$$\begin{aligned} \int x^3 e^x dx \\ &= x^3 e^x - 3x^2 e^x + 6x e^x - 6e^x + C. \end{aligned}$$

(')		(∫)
u		v
x ³	— +	e ^x
3x ²	— -	e ^x
6x	— +	e ^x
6	— -	e ^x
0	—	e ^x

註：以上設計必須有一多項式，如此微分夠多次會成 0 (如此 integration by parts 的過程才能停止)，最好是另一函數 (如 Example 5 的 e^x ，和 Example $\sin x$) 很容易找出反導數。

Example 6 : $\int x^2 \sin x dx.$

Solution :

$$\begin{aligned} \int x^2 \sin x dx \\ &= -x^2 \cos x + 2x \sin x + 2 \cos x + C. \end{aligned}$$

(')		(∫)
u		v
x ²	— +	sin x
2x	— -	-cos x
2	— +	-sin x
0	—	cos x

Example 7 : $\int e^x \sin x dx$.

Solution :

$$\begin{aligned} & \int e^x \sin x dx \\ &= e^x \sin x - \int e^x \cos x dx \\ &= e^x \sin x - e^x \cos x - \int e^x \sin x dx \\ \Rightarrow \int e^x \sin x dx &= \frac{1}{2} e^x (\sin x - \cos x) + C. \end{aligned}$$

$u = \sin x$	$dv = e^x dx$
$du = \cos x dx$	$v = e^x$
$u = \cos x$	$dv = e^x dx$
$du = -\sin x dx$	$v = e^x$

Example 8 : $\int e^{\sqrt{x}} dx$. (Hint : Substitution + Integration by Parts)

Solution :

$$u = \sqrt{x} \quad du = \frac{1}{2} x^{-\frac{1}{2}} dx = \frac{dx}{2u} \Rightarrow dx = 2u du$$

$$\begin{aligned} & \int e^{\sqrt{x}} dx \\ &= 2 \int u e^u du \\ &= 2u e^u - 2e^u + C \\ &= 2\sqrt{x} e^{\sqrt{x}} - 2e^{\sqrt{x}} + C. \end{aligned}$$

(')	(j)
u	e^u
1	e^u
0	e^u